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Ring-Zero

Ant Group Scales Zero RL to One Trillion Parameters, Observing Five Emergent Reasoning Behaviors and 84.2% on AIME 2026 Without Human-Annotated Data

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Zero reinforcement learning — RL with verifiable rewards and no human-annotated data — has demonstrated striking results at the 70B scale, but whether its training dynamics and emergent capabilities extend to far larger models remained unknown. Ant Group’s InclusionAI now closes that gap: Ring-Zero applies zero RL to Ring-2.5-1T, a one-trillion-parameter hybrid linear-attention model, and documents not only successful scaling but five qualitatively new behaviors that appear only at this scale.

AT A GLANCE

Problem: Naive scaling of zero RL collapses beyond ~70B parameters due to readability degradation, token redundancy, and distribution drift.

Why it matters: Trillion-parameter reasoning models are now practical; understanding and controlling zero RL at that scale is necessary for continued progress.

Method: Clipped importance sampling, training-inference ratio correction, and mixed-precision control applied to Ring-2.5-1T (1T parameters, 1M context window).

Result: 84.2% on AIME 2026 with first-stage training only; five new emergent behaviors including self-verification and context anxiety.

Evidence: Evaluated on AIME 2024/2025/2026, HMMT, and IMOAnswerBench; seven-benchmark average 74.8%.

The Big Picture

Zero RL uses only the binary signal of whether the model’s answer is correct to teach chain-of-thought reasoning — no annotated thought chains, no human preference data, no teacher demonstrations. DeepSeek-R1-Zero showed that this surprisingly minimal training signal is sufficient to produce sophisticated mathematical reasoning at the 70B scale, but every subsequent study has operated at or below that size.

The trillion-parameter regime matters because it is where qualitatively new capabilities have historically emerged: GPT-4-class reasoning, multimodal integration, and in-context meta-learning all appeared near or above the 1T parameter threshold. Ring-Zero asks whether the *training dynamics* of zero RL also exhibit threshold behavior at this scale — and finds that they do.

Why Existing Approaches Fall Short

Three failure modes obstruct naive trillion-parameter zero RL. First, **poor readability**: at large scale, unconstrained policy gradient updates cause the model to generate verbose, repetitive internal monologues that consume tokens without reasoning. Second, **token redundancy**: the model learns to pad reasoning traces with low-effort repetition, inflating apparent solution length while contributing no additional correctness. Third, **distribution drift**: at 1T parameters, rollout distributions shift rapidly relative to the behavior policy used for importance-weighted gradient estimation, destabilizing training after a few hundred steps.

Existing stabilization techniques developed for 70B models (gradient clipping, KL penalties) are insufficient at 1T because the optimizer’s effective learning rate — after importance-weighted corrections — can grow dramatically when policy entropy is high in a trillion-parameter space.

Core Mechanism

Ring-Zero addresses the three failure modes with three targeted interventions. **Clipped importance sampling** restricts the off-policy correction ratio to $[1 - \epsilon, 1 + \epsilon]$ per token, preventing any single high-variance rollout from dominating the gradient update. **Training-inference ratio correction** compensates for the mismatch between the truncated context windows used during RL rollouts and the full context available at inference. **Mixed-precision control** uses BF16 for activations and FP32 for gradient accumulation, with selective FP8 for embedding layers to keep the 1T model within GPU memory constraints.

Technical Formulation

The training-inference ratio correction scales the effective reward by the ratio of inference to training context lengths:

$$\hat{r}_t = r_t \cdot \left(\frac{C_{\text{inf}}}{C_{\text{train}}} \right)^\gamma$$

where C_{inf} is the inference context length, C_{train} is the rollout context length, and γ is a hyperparameter tuned per model size (set to 0.4 for Ring-2.5-1T).

Clipped importance sampling modifies the standard im-

portance weight:

$$\tilde{\rho}_t = \min\left(\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, 1 + \epsilon\right)$$

without a lower clip (since the lower clip would suppress on-policy updates when the policy has improved).

Learning Procedure

Training reliably unfolds in two sequential phases identified across multiple random seeds:

1. **Discovery phase** (first $\approx 30\%$ of steps): The model actively expands reasoning boundaries, activating dormant pathway combinations that were not used in pre-training. Output diversity is high; format is unstable.
2. **Sharpening phase** (remaining $\approx 70\%$ of steps): The model refines its policy within the stylistic and procedural boundaries established in discovery. Output quality and format consistency improve monotonically.

The phase transition is sharp and reproducible: gradient entropy drops by approximately 40% between steps 800 and 1200 across all runs, marking the boundary between discovery and sharpening.

Five Emergent Behaviors

Five behaviors appear at trillion-parameter scale that are absent at 70B, identified through systematic activation probing and output pattern analysis:

1. **Self-verification:** The model spontaneously checks intermediate conclusions before committing to a final answer, inserting verification sub-problems without being prompted to.
2. **Context anxiety:** The model models itself as a resource-constrained agent operating within a finite context window, making strategic decisions about what to include based on this self-model.
3. **Anthropomorphism:** The model adopts a persistent first-person problem-solving persona and maintains it across long reasoning chains.
4. **Structured formatting:** Headers, numbered steps, and boxed answers emerge without any formatting reward.
5. **Parallel reasoning:** The model explores multiple solution strategies in parallel within a single forward pass, then selects among them.

Experimental Findings

Benchmark	Ring-2.5-1T-Zero	Prior SOTA
AIME 2026	84.2%	78.5%
AIME 2025	81.7%	79.3%
AIME 2024	83.4%	82.1%
HMMT	78.3%	74.0%
IMOAnswerBench	61.2%	55.7%
7-Benchmark Avg.	74.8%	70.3%

Results are vendor-reported by InclusionAI; independent third-party evaluation was not yet available at press time.

Ablations and Interpretation

Without clipped importance sampling: Training diverges after approximately 500 steps at 1T scale; the effect is absent at 70B, confirming it is a trillion-parameter-specific failure mode.

Without training-inference ratio correction: AIME 2026 accuracy drops from 84.2% to 79.1%, a 5.1 percentage point loss attributable to underestimating long-context performance during training.

Context anxiety phase transition: The behavior is absent below $\sim 800\text{B}$ parameters and strengthens above 1T, suggesting a phase transition rather than a smooth scaling curve.

Self-verification contribution: Suppressing self-verification tokens at inference reduces AIME 2026 accuracy by approximately 2–3 percentage points, confirming its functional role in correctness.

Reference

Xinyu Tang et al. **Ring-Zero: Scaling Zero RL to a Trillion Parameters for Emergent Reasoning**. arXiv:2607.12395, July 2026. <https://arxiv.org/abs/2607.12395>